

The greenhouse brain

For the controller in the scene, or one like it, the open-source project Mycodo turns a Raspberry Pi into an environmental controller: sensors in, relays out, and feedback loops in between.

- **Sensors:** an air temperature and humidity sensor (the SHT31-D is common) and a capacitive soil-moisture probe, which resists corrosion better than the cheap resistive kind.
- **Actuators:** a relay board switching low-voltage devices: an exhaust fan, a solenoid valve on the irrigation line, supplemental lights. Have someone who knows electrical work check anything that touches mains power.

On a clean Raspberry Pi OS install, check the project's current installation requirements, then use its documented installer:

```
sudo apt update && sudo apt upgrade -y
curl -L https://kizniche.github.io/Mycodo/install | bash
```

Then browse to <https://<your-pi-ip>/> for the web interface. From there you can set up PID (proportional-integral-derivative) loops, the standard control law for holding a value steady:

$$u(t) = K_p e(t) + K_i \int e(\tau) d\tau + K_d de(t)/dt$$

where $e(t)$ is the gap between the target and the sensor reading, and the three K values tune the response. A valve that only opens or closes still needs limits on run time and total water, plus an off state if the sensor stops reporting. Start with a grower's judgment about what the plants need, keep the manual valve working, and check the soil with your hands until you trust the numbers. (Chapter 15 is about what happens when you don't.)